

THE MINUM'S SYSTEM MANAGEMENT APP (TMS).

COLLABORATION BETWEEN MEGAMIND AND THE MINUM'S ENT.

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ABOUT US

Development Team:
MegaMind (Group 04)

Team Motto:
Smart Solutions, Seamless Experiences.

Core Solution:
A cross-platform mobile application built with Flutter and integrated with AI customer chatbot.

GROUP MEMBERS:



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INTRODUCTION

Our team proposes a comprehensive digital ordering and operational hub for THE MINUM'S ENT. We aim to transition customer interactions from traditional face-to-face service to a seamless mobile platform. The application is designed to streamline the entire ordering process, serving as a central bridge between customers and staff. It will feature an AI-powered chatbot that assists customers by answering questions about menu items, prices, and product availability, providing a more convenient and efficient ordering experience. If the chatbot cannot answer a question, it will direct customers to contact the manager for further assistance. Additionally, a real-time inventory management system will help staff monitor product status and track material availability, supporting smoother daily operations and more effective inventory control.

CLIENT BACKGROUND

THE MINUM'S™
FRESH • SWEET • FLAVOURFUL

Established in 2024, THE MINUM'S ENT. is a food and beverage enterprise focused on delivering quality refreshments and customer experiences. The project's primary client contact is the founder, Petrus Anak Jinai. The proposed digital ordering application and operational hub are designed to support an estimated 100 intended users, encompassing both the business's operational staff and its daily customer base.

PROJECT OBJECTIVES

1. To study and analyze THE MINUM'S ENT's operational workflows, focusing primarily on staff attendance tracking and inventory management, followed by customer order handling
2. To design and develop the core Flutter mobile app featuring digital staff check-in/check-out tracking and real-time inventory stock management.
3. To integrate a Cloud Generative AI module (Google Gemini) to assist with menu queries, availability, and inventory stock insights.
4. To implement digital customer ordering functionality and evaluate the complete system with the client through User Acceptance Testing (UAT).

PROBLEM STATEMENT

- P1: **Manual Staff Attendance & Work Log**
  - Difficulty in tracking staff check-in/check-out, working hours, and staff availability, resulting in high friction in monitoring operations.
- P2: **Manual Inventory Tracking**
  - High risk of stock shortages or overstocking due to reliance on paper/manual record-keeping, complicating ingredient level monitoring.
- P3: **Manual Customer Ordering**
  - Risk of missed or miscommunicated orders during peak hours, leading to customer service delays and queue bottlenecks.

PROPOSED SOLUTION

Similar problems have been observed in manual restaurant ordering processes, where staff handling multiple orders can result in late, incorrect, unclear, or forgotten orders (Wong & Kamaludin, 2023). A Flutter mobile app that manages staff attendance, inventory, and customer orders in one centralized system, using AI to provide useful insights.

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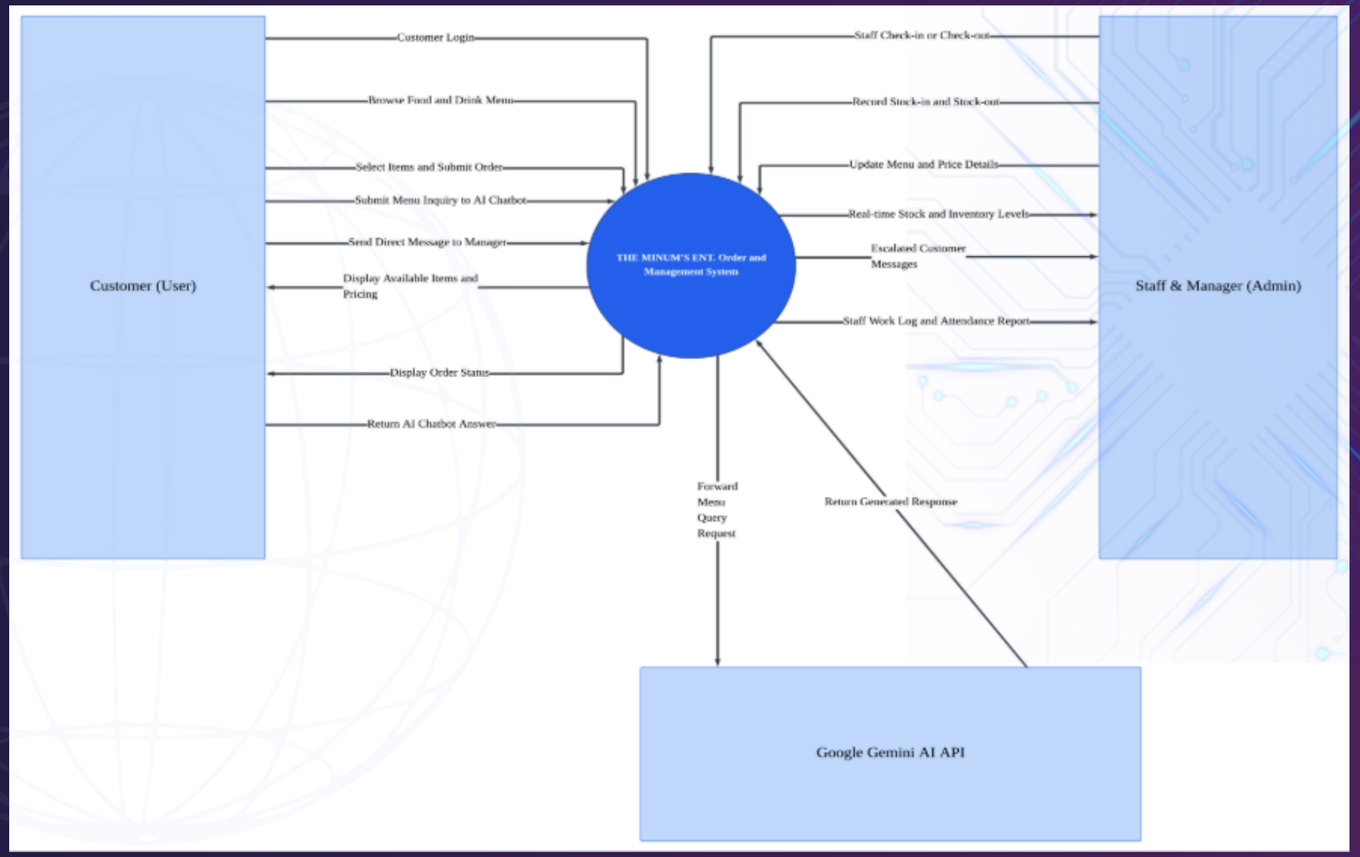
TARGET USERS

User type	Description	Main needs / goals
Customer	Daily visitors that purchasing food and beverage from THE MINUM	• Browse menus, check live pricing and availability of the product. • Interact with AI chatbox for quick input and services. • Place order directly.
Staff	Operational workers that handling service, order fulfillment, and inventory stock tracking.	• Perform digital check-in or check-out attendance tracking • Directly record stock-in or stock-out transaction • Update product availibiliy
Manager/CEO	The business founder and administrator that manage overall operations, staff hours, and inventory	• Monitor staff attendance records and working hours • Manage menus items, prices, and inventory levels. • Review customers questions and feedback escalated by AI chatbo

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SYSTEM CONTEXT AND ARCHITECTURE

THE SYSTEM CONTEXT DIAGRAM FOR THE MINUM'S ENT. MANAGEMENT SYSTEM SHOWS HOW THE SYSTEM INTERACTS WITH CUSTOMER, STAFF & MANAGER, AND GOOGLE GEMINI AI API AT THE HIGHER LEVEL. THE CUSTOMERS ARE ABLE TO LOG INTO THE MOBILE APP TO VIEW THE MENU, PLACE AN ORDER, TRACK THEIR ORDER ONLINE, AND ALSO SEND INQUIRIES THROUGH THE AI CHATBOT OR DIRECTLY TO MANAGERS. STAFF & MANAGER WILL BE INTERACTING WITH THE SYSTEM TO TAKE CARE OF THEIR ATTENDANCE, REGISTER STOCK-IN AND STOCK-OUT, EDIT MENU INFORMATION, AND VIEW REPORTS RELATED TO INVENTORY AND WORK LOG.



CONTEXT DIAGRAM OF THE MINUM ENT. MANAGEMENT SYSTEM

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A.I FEATURE

Item	Description
AI capability	AI chatbot for answering customer questions about the menu.
User value	Helps customers quickly check food and drink prices, availability and other menu information. It also allows customers to contact the manager if the AI cannot answer.
Input → Output	Input: Customer's question and menu information. Output: An AI-generated answer based on the available menu information.
AI service / API	Google Gemini API.
Constraints	Protect the API key, control API usage costs, handle slow responses and prevent the AI from providing incorrect menu information. Customers can contact the manager if the AI cannot answer.

TASK ALLOCATION & TEAM CONTRIBUTION

Member	TSPi role	Contribution
Caleb	Software Project Manager (PM)	Overall project scheduling, sprint planning, risk escalation, GitHub Projects board administration, compiling weekly reports (WEEK/SCHEDULE), and team coordination.
Fahdil	Lead Mobile Architect (UI/UX)	UI/UX prototyping, Provider state management architecture, Flutter Material 3 design system, database schema design, and modular Dart architecture.
Joeshua	Lead Systems Analyst & Requirements Engineer	Client requirements elicitation, authoring SRS and PRD, requirement traceability matrix, managing client communication/meeting evidence, and UAT coordination.
Reynoldson	Backend & AI Integration Specialist	Backend API and Firebase database integration, Google Gemini AI integration, AI error/fallback handling, maintaining AI audit log (LOGAI), and PR code reviews.
Matthias	Quality Assurance & CI/CD Engineer	GitHub Actions CI/CD workflow, automated unit/widget test planning, defect tracking (LOGD), maintaining test execution log (LOGTEST), ensuring ≥60% code coverage, and APK release packaging.

EXPECTED RESULTS

The final deliverable of this project is **THE MINUM'S ENT. System Management Application (v1.0.0)**, a fully functional cross-platform Android application that integrates customer ordering, staff attendance management, and inventory tracking into a single platform. Customers can register, log in, browse the food and beverage menu, check item availability, and place orders through the application. The application also includes a **Google Gemini-powered AI chatbot** to answer customer enquiries about menu prices, item availability, and kiosk services, with an option to contact the manager for further assistance when needed.

Staff and administrators can access role-based features to manage attendance records, monitor inventory levels, and record stock movements. These features aim to improve daily operations, maintain accurate records, and streamline kiosk management.

The project will be considered successfully completed when the release APK (v1.0.0) has been tested, at least 50 test cases have been executed, code coverage reaches a minimum of 60%, and no critical defects remain unresolved in the Defect Log (LOGD). Formal client approval must also be obtained through the UAT Acceptance Form (FORM-CLIENT-ACCEPT) by Milestone 4 (Week 15).

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THANK YOU

THANK YOU FOR YOUR ATTENTION!
LET’S SHAPE THE FUTURE TOGETHER